Solution Equations (2.176) are also analogous to electrical node equations. Coefficients represent sums of electrical admittances. Admittances associated with M_1 form the elements connected to the first node, where mechanical admittances between the two masses are common to the two nodes. Mechanical admittances associated with M_2 form the elements connected to the second node. The result is shown in Figure 2.39, where $v_1(t)$ and $v_2(t)$ are the velocities of M_1 and M_2, respectively.

2.10 Nonlinearities

The models thus far are developed from systems that can be described approximately by linear, time-invariant differential equations. An assumption of *linearity* was implicit in the development of these models. In this section, we formally define the terms *linear* and *nonlinear* and show how to distinguish between the two. In Section 2.11 we show how to approximate a nonlinear system as a linear system, so that we can use the modeling techniques previously covered in this chapter (Hsu, 1968).

A linear system possesses two properties: superposition and homogeneity. The property of *superposition* means that the output response of a system to the sum of inputs is the sum of the responses to the individual inputs. Thus, if an input of $r_1(t)$ yields an output of $c_1(t)$ and an input of $r_2(t)$ yields an output of $c_2(t)$, then an input of $r_1(t) + r_2(t)$ yields an output of $c_1(t) + c_2(t)$. The property of *homogeneity* describes the response of the system to a multiplication of the input by a scalar. Specifically, in a linear system, the property of homogeneity is demonstrated if for an input of $r_1(t)$ that yields an output of $c_1(t)$, an input of $Ar_1(t)$ yields an output of $Ac_1(t)$; that is, multiplication of an input by a scalar yields a response that is multiplied by the same scalar.

We can visualize linearity as shown in Figure 2.40. Figure 2.40(*a*) is a linear system where the output is always 1/2 the input, or $f(x) = 0.5x$, regardless of the value of x. Thus each of the two properties of linear systems applies. For example, an input of 1 yields an output of 1/2 and an input of 2 yields an output of 1. Using superposition, an input that is the sum of the original inputs, or 3, should yield an output that is the sum of the individual outputs, or 1.5. From Figure 2.40(*a*), an input of 3 does indeed yield an output of 1.5.

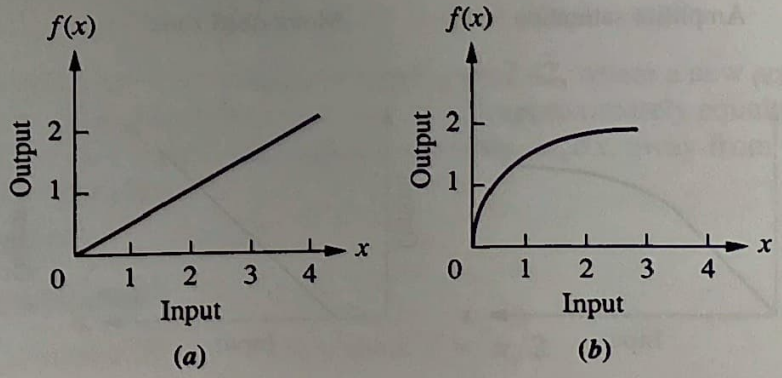


Figure 2.40
a. Linear system;
b. nonlinear system

To test the property of homogeneity, assume an input of 2, which yields an output of 1. Multiplying this input by 2 should yield an output of twice as much, or 2. From Figure 2.40(*a*), an input of 4 does indeed yield an output of 2. The reader can verify that the properties of linearity certainly do not apply to the relationship shown in Figure 2.40(*b*).

Figure 2.41 shows some examples of physical nonlinearities. An electronic amplifier is linear over a specific range but exhibits the nonlinearity called *saturation* at high input voltages. A motor that does not respond at very low input voltages due to frictional forces exhibits a nonlinearity called *dead zone*. Gears that do not fit tightly exhibit a nonlinearity called *backlash:* the input moves over a small range without the output responding. The reader should verify that the curves shown in Figure 2.41 do not fit the definitions of linearity over their entire range. Another example of a nonlinear subsystem is a phase detector, used in a phase-locked loop in an FM radio receiver, whose output response is the sine of the input.

A designer can often make a linear approximation to a nonlinear system. Linear approximations simplify the analysis and design of a system and are used as long as the results yield a good approximation to reality. For example, a linear relationship can be established at a point on the nonlinear curve if the range of input values about that point is small and the origin is translated to that point. Electronic amplifiers are an example of physical devices that perform linear amplification with small excursions about a point.

2.11 Linearization

The electrical and mechanical systems covered thus far were assumed to be linear. However, if any nonlinear components are present, we must linearize the system before we can find the transfer function. In the last section we defined and discussed nonlinearities; in this section we show how to obtain linear approximations to nonlinear systems in order to obtain transfer functions.

The first step is to recognize the nonlinear component and write the nonlinear differential equation. When we linearize a nonlinear differential equation, we linearize it for small-signal inputs about the steady-state solution when the small-signal input is equal to zero. This steady-state solution is called *equilibrium* and is selected as the second step in the linearization process. For example, when a

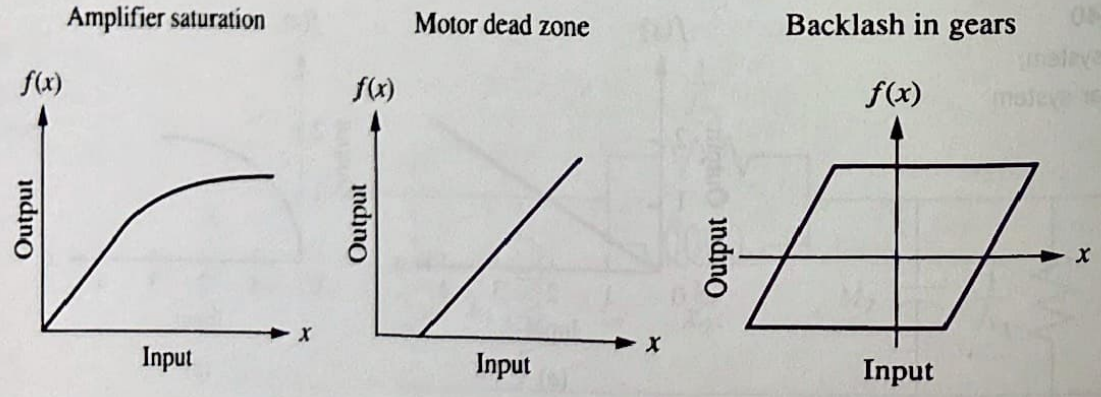


Figure 2.41 Some physical nonlinearities

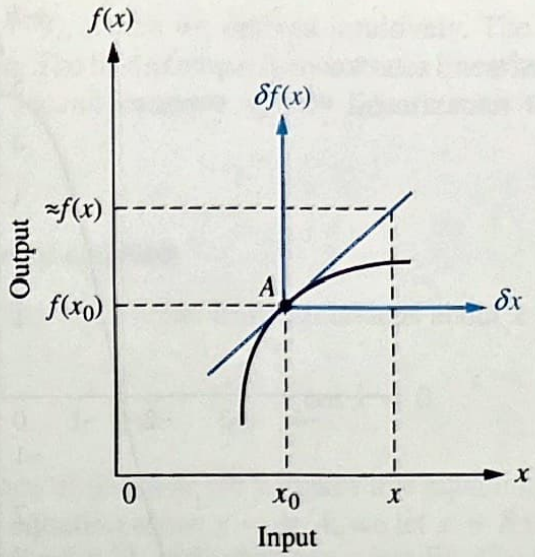


Figure 2.42
Linearization about a point *A*

pendulum is at rest, it is at equilibrium. The angular displacement is described by a nonlinear differential equation, but it can be expressed with a linear differential equation for small excursions about this equilibrium point.

Next, we linearize the nonlinear differential equation, and then we take the Laplace transform of the linearized differential equation, assuming zero initial conditions. Finally, we separate input and output variables and form the transfer function. Let us first see how to linearize a function; later, we will apply the method to the linearization of a differential equation.

If we assume a nonlinear system operating at point A, $[x_0, f(x_0)]$ in Figure 2.42, small changes in the input can be related to changes in the output about the point by way of the slope of the curve at the point A. Thus, if the slope of the curve at point A is m_a, then small excursions of the input about point A, δx, yield small changes in the output, $\delta f(x)$, related by the slope at point A. Thus,

$$[f(x) - f(x_0)] \approx m_a(x - x_0) \tag{2.178}$$

from which

$$\delta f(x) \approx m_a \delta x \tag{2.179}$$

and

$$f(x) \approx f(x_0) + m_a(x - x_0) \approx f(x_0) + m_a \delta x \tag{2.180}$$

This relationship is shown graphically in Figure 2.42, where a new set of axes, δx and $\delta f(x)$, is created at the point A, and $f(x)$ is approximately equal to $f(x_0)$, the ordinate of the new origin, plus small excursions, $m_a \delta x$, away from point A. Let us look at an example.

Example 2.26

Linearizing a function

Problem Linearize $f(x) = 5\cos x$ about $x = \pi/2$.

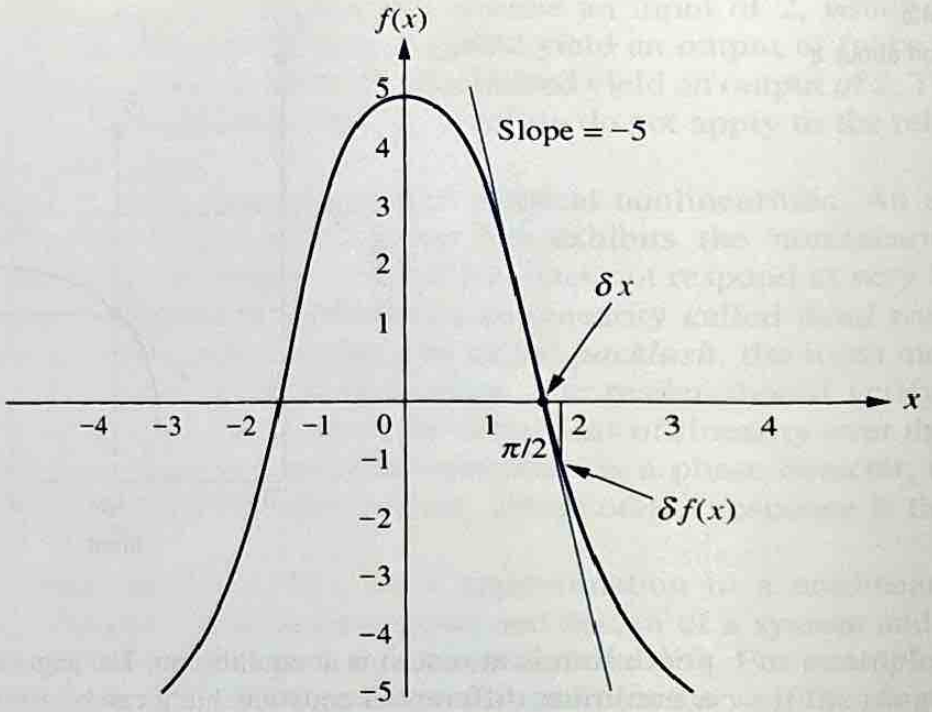


Figure 2.43
Linearization of $5\cos x$ about $x = \pi/2$

Solution We first find that the derivative of $f(x)$ is $df/dx = (-5\sin x)$. At $x = \pi/2$, the derivative is -5. Also $f(x_0) = f(\pi/2) = 5\cos(\pi/2) = 0$. Thus, from Eq. (2.180), the system can be represented as $f(x) = -5\,\delta x$ for small excursions of x about $\pi/2$. The process is shown graphically in Figure 2.43, where the cosine curve does indeed look like a straight line of slope -5 near $\pi/2$.

The previous discussion can be formalized using the Taylor series expansion, which expresses the value of a function in terms of the value of that function at a particular point, the excursion away from that point, and derivatives evaluated at that point. The Taylor series is shown in Eq. (2.181).

$$f(x) = f(x_0) + \left.\frac{df}{dx}\right|_{x=x_0} \frac{(x-x_0)}{1!} + \left.\frac{d^2f}{dx^2}\right|_{x=x_0} \frac{(x-x_0)^2}{2!} + \cdots \tag{2.181}$$

For small excursions of x from x_0, we can neglect higher-order terms. The resulting approximation yields a straight-line relationship between the change in $f(x)$ and the excursions away from x_0. Neglecting the higher-order terms in Eq. (2.181), we get

$$f(x) - f(x_0) \approx \left.\frac{df}{dx}\right|_{x=x_0} (x - x_0) \tag{2.182}$$

or

$$\delta f(x) \approx m|_{x=x_0}\,\delta x \tag{2.183}$$

which is a linear relationship between $\delta f(x)$ and δx for small excursions away from x_0. It is interesting to note that Eqs. (2.182) and (2.183) are identical to

Eqs. (2.178) and (2.179), which we derived intuitively. The following examples illustrate linearization. The first example demonstrates linearization of a differential equation, while the second example applies linearization to finding a transfer function.

Example 2.27

Linearizing a differential equation

Problem Linearize Eq. (2.184) for small excursions about $x = \pi/4$.

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + \cos x = 0 \tag{2.184}$$

Solution The presence of the term $\cos x$ makes this equation nonlinear. Since we want to linearize the equation about $x = \pi/4$, we let $x = \delta x + \pi/4$, where δx is the small excursion about $\pi/4$, and substitute x into Eq. (2.184):

$$\frac{d^2\left(\delta x + \frac{\pi}{4}\right)}{dt^2} + 2\frac{d\left(\delta x + \frac{\pi}{4}\right)}{dt} + \cos\left(\delta x + \frac{\pi}{4}\right) = 0 \tag{2.185}$$

But

$$\frac{d^2\left(\delta x + \frac{\pi}{4}\right)}{dt^2} = \frac{d^2\delta x}{dt^2} \tag{2.186}$$

and

$$\frac{d\left(\delta x + \frac{\pi}{4}\right)}{dt} = \frac{d\delta x}{dt} \tag{2.187}$$

Finally, the term $\cos(\delta x + (\pi/4))$ can be linearized with the truncated Taylor series. Substituting $f(x) = \cos(\delta x + (\pi/4))$, $f(x_0) = f(\pi/4) = \cos(\pi/4)$, and $(x - x_0) = \delta x$ into Eq. (2.182) yields

$$\cos\left(\delta x + \frac{\pi}{4}\right) - \cos\left(\frac{\pi}{4}\right) = \left.\frac{d\cos x}{dx}\right|_{x=\frac{\pi}{4}} \delta x = -\sin\left(\frac{\pi}{4}\right)\delta x \tag{2.188}$$

Solving Eq. (2.188) for $\cos(\delta x + (\pi/4))$, we get

$$\cos\left(\delta x + \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)\delta x = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\delta x \tag{2.189}$$

Substituting Eqs. (2.186), (2.187), and (2.189) into Eq. (2.185) yields the following linearized differential equation:

$$\frac{d^2\delta x}{dt^2} + 2\frac{d\delta x}{dt} - \frac{\sqrt{2}}{2}\delta x = -\frac{\sqrt{2}}{2} \tag{2.190}$$

This equation can now be solved for δx, from which we can obtain $x = \delta x + (\pi/4)$.

Even though the nonlinear Eq. (2.184) is homogeneous, the linearized Eq. (2.190) is not homogeneous. Eq. (2.190) has a forcing function on its right-hand side. This additional term can be thought of as an input to a system represented by Eq. (2.184).

Another observation about Eq. (2.190) is the negative sign on the left-hand side. The study of differential equations tells us that since the roots of the characteristic equation are positive, the homogeneous solution grows without bound instead of diminishing to zero. Thus, this system linearized around $x = \pi/4$ is not stable.

Example 2.28

Transfer function—nonlinear electrical network

Problem Find the transfer function, $V_L(s)/V(s)$, for the electrical network shown in Figure 2.44, which contains a nonlinear resistor whose voltage-current relationship is defined by $i_r = 2e^{0.1v_r}$, where i_r and v_r are the resistor current and voltage, respectively. Also, $v(t)$ in Figure 2.44 is a small-signal source.

Solution We will use Kirchhoff's voltage law to sum the voltages in the loop to obtain the nonlinear differential equation, but first we must solve for the voltage across the nonlinear resistor. Taking the natural log of the resistor's current-voltage relationship, we get $v_r = 10 \ln \frac{1}{2} i_r$. Applying Kirchhoff's voltage law around the loop, where $i_r = i$, yields

$$\frac{di}{dt} + 10 \ln \frac{1}{2} i - 20 = v(t) \tag{2.191}$$

Next, let us evaluate the equilibrium solution. First, set the small-signal source, $v(t)$, equal to zero. Now evaluate the steady-state current. With $v(t) = 0$, the circuit consists of a 20 V battery in series with the inductor and nonlinear resistor. In the steady state, the voltage across the inductor will be zero, since $v_L(t) = L\,di/dt$ and di/dt is zero in the steady state, given a constant battery source. Hence, the resistor voltage, v_r, is 20 V. Using the characteristics of the resistor, $i_r = 2e^{0.1v_r}$,

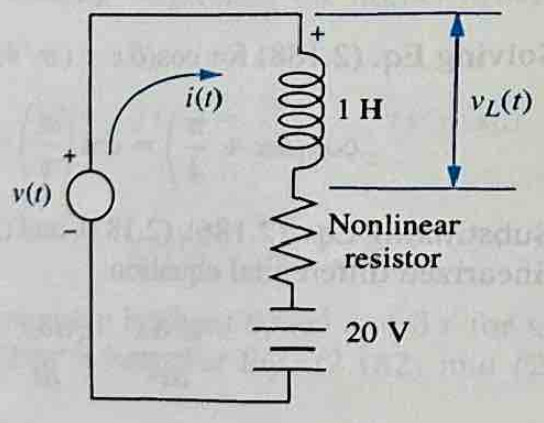


Figure 2.44
Nonlinear electrical network

we find that $i_r = i = 14.78$ amps. This current, i_0, is the equilibrium value of the network current. Hence $i = i_0 + \delta i$. Substituting this current into Eq. (2.191) yields

$$L\frac{d(i_0 + \delta i)}{dt} + 10\ln\frac{1}{2}(i_0 + \delta i) - 20 = v(t) \tag{2.192}$$

Using Eq. (2.182) to linearize $\ln\frac{1}{2}(i_0 + \delta i)$, we get

$$\ln\frac{1}{2}(i_0 + \delta i) - \ln\frac{1}{2}i_0 = \left.\frac{d(\ln\frac{1}{2}i)}{di}\right|_{i=i_0} \delta i = \left.\frac{1}{i}\right|_{i=i_0} \delta i = \frac{1}{i_0}\delta i \tag{2.193}$$

or

$$\ln\frac{1}{2}(i_0 + \delta i) = \ln\frac{i_0}{2} + \frac{1}{i_0}\delta i \tag{2.194}$$

Substituting into Eq. (2.192), the linearized equation becomes

$$L\frac{d\delta i}{dt} + 10\left(\ln\frac{i_0}{2} + \frac{1}{i_0}\delta i\right) - 20 = v(t) \tag{2.195}$$

Letting $L = 1$ and $i_0 = 14.78$, the final linearized differential equation is

$$\frac{d\delta i}{dt} + 0.677\delta i = v(t) \tag{2.196}$$

Taking the Laplace transform with zero initial conditions and solving for $\delta i(s)$, we get

$$\delta i(s) = \frac{V(s)}{s + 0.677} \tag{2.197}$$

But the voltage across the inductor about the equilibrium point is

$$v_L(t) = L\frac{d}{dt}(i_0 + \delta i) = L\frac{d\delta i}{dt} \tag{2.198}$$

Taking the Laplace transform,

$$V_L(s) = Ls\delta i(s) = s\delta i(s) \tag{2.199}$$

Substituting Eq. (2.197) into Eq. (2.199) yields

$$V_L(s) = s\frac{V(s)}{s + 0.677} \tag{2.200}$$

from which the final transfer function is

$$\frac{V_L(s)}{V(s)} = \frac{s}{s + 0.677} \tag{2.201}$$

for small excursions about $i = 14.78$ or, equivalently, about $v(t) = 0$.

Case Studies

Antenna Control: Transfer Functions

This chapter showed that physical systems can be modeled mathematically with transfer functions. Typically, systems are composed of subsystems of different types, such as electrical, mechanical, and electromechanical.

The first case study uses our ongoing example of the antenna azimuth position control system to show how to represent each subsystem as a transfer function.

Problem Find the transfer function for each subsystem of the antenna azimuth position control system schematic shown on the inside front cover. Use Configuration 1.

Solution First, we identify the individual subsystems for which we must find transfer functions; they are summarized in Table 2.6. We proceed to find the transfer function for each subsystem.

Input Potentiometer; Output Potentiometer Since the input and output potentiometers are configured in the same way, their transfer functions will be the same. We *neglect* the dynamics for the potentiometers and simply find the relationship between the output voltage and the input angular displacement. In the center position, the output voltage is zero. Five turns toward either the positive 10 volts or the negative 10 volts yields a voltage change of 10 volts. Thus, the transfer function, $V_i(s)/\theta_i(s)$, for the potentiometers is found by dividing the voltage change by the angular displacement:

$$\frac{V_i(s)}{\theta_i(s)} = \frac{10}{10\pi} = \frac{1}{\pi} \tag{2.202}$$

Preamplifier; Power Amplifier The transfer functions of the amplifiers are given in the problem statement. Two phenomena are *neglected.* First, we *assume* that saturation is never reached. Second, the dynamics of the preamplifier

Table 2.6 Subsystems of the antenna azimuth position control system

Subsystem	Input	Output
Input potentiometer	Angular rotation from user $\theta_i(t)$	Voltage to preamp $v_i(t)$
Preamp	Voltage from potentiometers $v_e(t) = v_i(t) - v_o(t)$	Voltage to power amp $v_p(t)$
Power amp	Voltage from preamp $v_p(t)$	Voltage to motor $e_a(t)$
Motor	Voltage from power amp $e_a(t)$	Angular rotation to load $\theta_0(t)$
Output potentiometer	Angular rotation from load $\theta_0(t)$	Voltage to preamp $v_0(t)$

are *neglected,* since its speed of response is typically much greater than that of the power amplifier. The transfer functions of both amplifiers are given in the problem statement and are the ratio of the Laplace transforms of the output voltage divided by the input voltage. Hence, for the preamplifier,

$$\frac{V_p(s)}{V_e(s)} = K \tag{2.203}$$

and for the power amplifier,

$$\frac{E_a(s)}{V_p(s)} = \frac{100}{s + 100} \tag{2.204}$$

Motor and Load The motor and its load are next. The transfer function relating the armature displacement to the armature voltage is given in Eq. (2.153). The equivalent inertia, J_m, is

$$J_m = J_a + J_L\left(\frac{25}{250}\right)^2 = 0.02 + 1\frac{1}{100} = 0.03 \tag{2.205}$$

where $J_L = 1$ is the load inertia at θ_0. The equivalent viscous damping, D_m, at the armature is

$$D_m = D_a + D_L\left(\frac{25}{250}\right)^2 = 0.01 + 1\frac{1}{100} = 0.02 \tag{2.206}$$

where D_L is the load viscous damping at θ_0. From the problem statement, $K_t = 0.5$ N-m/A, $K_b = 0.5$ V-s/rad, and the armature resistance $R_a = 8$ ohms. These quantities along with J_m and D_m are substituted into Eq. (2.153), yielding the transfer function of the motor from the armature voltage to the armature displacement, or

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K_t/(R_aJ_m)}{s\left[s + \frac{1}{J_m}\left(D_m + \frac{K_tK_b}{R_a}\right)\right]} = \frac{2.083}{s(s + 1.71)} \tag{2.207}$$

To complete the transfer function of the motor, we multiply by the gear ratio to arrive at the transfer function relating load displacement to armature voltage:

$$\frac{\theta_0(s)}{E_a(s)} = 0.1\frac{\theta_m(s)}{E_a(s)} = \frac{0.2083}{s(s + 1.71)} \tag{2.208}$$

The results are summarized in the block diagram and table of block diagram parameters (Configuration 1) shown on the inside front cover.

Challenge We now give you a problem to test your knowledge of this chapter's objectives: Referring to the antenna azimuth position control system schematic shown on the inside front cover, evaluate the transfer function of each subsystem. Use Configuration 2. Record your results in the table of block diagram parameters for use in subsequent chapters' case study challenges.

Transfer Function of a Human Leg

In this case study, we find the transfer function of a biological system. The system is a human leg, which pivots from the hip joint. In this problem the component of weight is nonlinear, so the system requires linearization before the evaluation of the transfer function.

Problem The transfer function of a human leg relates the output angular rotation about the hip joint to the input torque supplied by the leg muscle. A simplified model for the leg is shown in Figure 2.45. The model *assumes* an applied muscular torque, $T_m(t)$, viscous damping, D, at the hip joint, and inertia, J, around the hip joint.[15] Also, a component of the weight of the leg, Mg, where M is the mass of the leg and g is the acceleration due to gravity, creates a nonlinear torque. If we *assume* that the leg is of uniform density, the weight can be applied at $L/2$, where L is the length of the leg (Milsum, 1966). Do the following:

a. Evaluate the nonlinear torque.

b. Find the transfer function, $\theta(s)/T_m(s)$, for small angles of rotation, where $\theta(s)$ is the angular rotation of the leg about the hip joint.

Solution First, calculate the torque due to the weight. The total weight of the leg is Mg acting vertically. The component of the weight in the direction of rotation is $Mg\sin\theta$. This force is applied at a distance $L/2$ from the hip joint. Hence the torque in the direction of rotation, $T_W(t)$, is $Mg(L/2)\sin\theta$. Next, draw a free-body diagram of the leg, showing the applied torque, $T_m(t)$, the torque due to the weight, $T_W(t)$, and the opposing torques due to inertia and viscous damping (see Figure 2.46).

Summing torques, we get

$$J\frac{d^2\theta}{dt^2} + D\frac{d\theta}{dt} + Mg\frac{L}{2}\sin\theta = T_m(t) \tag{2.209}$$

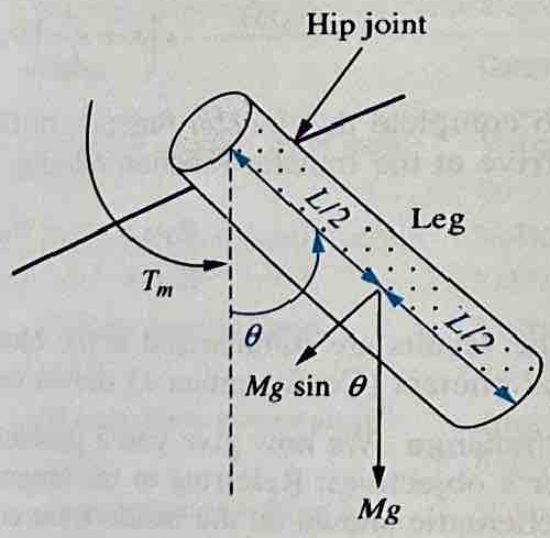


Figure 2.45
Cylinder model of a human leg

[15]For emphasis, J is not around the center of mass, as we previously assumed for inertia in mechanical rotation.

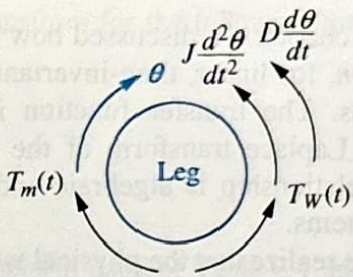


Figure 2.46 Free-body diagram of leg model

We linearize the system about the equilibrium point, $\theta = 0$, the vertical position of the leg. Using Eq. (2.182), we get

$$\sin\theta - \sin 0 = (\cos 0)\,\delta\theta \tag{2.210}$$

from which, $\sin\theta = \delta\theta$. Also, $J\,d^2\theta/dt^2 = J\,d^2\delta\theta/dt^2$ and $D\,d\theta/dt = D\,d\delta\theta/dt$. Hence Eq. (2.209) becomes

$$J\frac{d^2\delta\theta}{dt^2} + D\frac{d\delta\theta}{dt} + Mg\frac{L}{2}\delta\theta = T_m(t) \tag{2.211}$$

Notice that the torque due to the weight approximates a spring torque on the leg. Taking the Laplace transform with zero initial conditions yields

$$\left(Js^2 + Ds + Mg\frac{L}{2}\right)\delta\theta(s) = T_m(s) \tag{2.212}$$

from which the transfer function is

$$\frac{\delta\theta(s)}{T_m(s)} = \frac{1/J}{s^2 + \frac{D}{J}s + \frac{MgL}{2J}} \tag{2.213}$$

for small excursions about the equilibrium point, $\theta = 0$.

Challenge We now introduce a case study challenge to test your knowledge of this chapter's objectives. Although the physical system is different from a human leg, the problem demonstrates the same principles—linearization followed by transfer function evaluation.

Given the nonlinear electrical network shown in Figure 2.47, find the transfer function relating the output nonlinear resistor voltage, $V_r(s)$, to the input source voltage, $V(s)$.

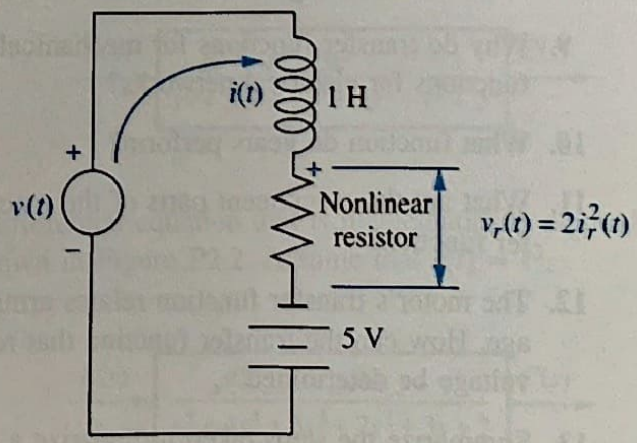


Figure 2.47 Nonlinear electric circuit